Probability Midterm 1

50 minutes • 100 points • Six problems

Grading note. Award method credit after an earlier error; do not deduct twice for the same mistake. Suggested working times for Problems 1–6 are 5, 5, 6, 8, 10, and 9 minutes, leaving 7 minutes to check.
Instructions. Answer all six problems and show your reasoning. Exact answers are acceptable; binomial coefficients need not be evaluated. No calculator is needed. Use the spaces provided; clearly label any work continued on the back. In Problem 6, use the guided steps rather than quoting a coupon-collector formula. Reserve time to check your work.

1. Sets — 10 points

Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, and $B = \{1, 2, 3, 4\}$. Take all complements relative to U.

(a) Find $A \cap B$, $A \setminus B$, and $(A \cup B)^c$. (6 points)

Solution.

$$A \cap B = \{2, 4\}, \qquad A \setminus B = \{6, 8, 10\}, \qquad (A \cup B)^c = \{5, 7, 9\}.$$

Credit: 2 points per correct set; allow partial credit for a single entry error.

(b) For these sets A, B, is the following statement true for every $C \subseteq U$? Justify your answer with a proof or a counterexample. (4 points)

$$C \subseteq A \cup B \quad \Longrightarrow \quad C \subseteq A \text{ or } C \subseteq B.$$

Solution. False. Take $C = \{1, 6\}$. Both elements lie in $A \cup B$, so $C \subseteq A \cup B$. But $1 \notin A$ and $6 \notin B$; therefore $C \not\subseteq A$ and $C \not\subseteq B$.

Credit: False: 1; valid counterexample: 1; verifying containment in the union: 1; verifying both failed containments: 1. Accept any valid counterexample.

2. A probability space — 10 points

Let $\Omega = \{a, b, c, d\}$, with every subset of Ω an event. The probabilities of the individual outcomes a, b, c, d are $k, 2k, 3k, 4k$, respectively.

(a) Find k, explaining the probability axiom you use. (4 points)

Solution. The disjoint singleton events partition Ω, and $\mathbb{P}(\Omega) = 1$. Finite additivity gives $k + 2k + 3k + 4k = 1$, so $k = 1/10$. All four probabilities are nonnegative.

Credit: Normalization/additivity explanation: 2; equation: 1; $k = 1/10$: 1.

(b) Let $E = \{a, c\}$ and $F = \{b, c\}$. Find $\mathbb{P}(E)$, $\mathbb{P}(E \cap F)$, and $\mathbb{P}(E^c)$. (6 points)

Solution. Since $E \cap F = \{c\}$ and $E^c = \{b, d\}$,

$$\mathbb{P}(E) = \frac{1+3}{10} = \frac{2}{5}, \qquad \mathbb{P}(E \cap F) = \frac{3}{10}, \qquad \mathbb{P}(E^c) = \frac{2+4}{10} = \frac{3}{5}.$$

Credit: 2 points per probability. If k was incorrect in (a), award full method credit for $4k, 3k, 6k$, or a consistently justified complement.

3. Inclusion–exclusion **15 points**

In a class of 40 students, 18 study French, 16 Spanish, and 12 German. Seven study both French and Spanish, five both French and German, and four both Spanish and German. Two study all three languages. **The pairwise counts include students who study all three.**

(a) How many students study at least one of these languages? Show your inclusion–exclusion calculation. (11 points)

Solution. Writing F, S, G for the three sets, inclusion–exclusion gives

$$\begin{aligned}|F \cup S \cup G| &= |F| + |S| + |G| - |F \cap S| - |F \cap G| - |S \cap G| + |F \cap S \cap G| \\ &= 18 + 16 + 12 - 7 - 5 - 4 + 2 = 32.\end{aligned}$$

Credit: Adding the three individual counts: 3; subtracting each pairwise intersection: 3; restoring the triple intersection once: 3; result 32: 2.

(b) How many students study none of these languages? (4 points)

Solution. There are $40 - |F \cup S \cup G| = 40 - 32 = 8$ such students.

Credit: Complement subtraction: 3; answer 8: 1. Carry forward an earlier error.

4. Balls in an urn **20 points**

An urn contains five red, four blue, and three green balls. Four balls are selected uniformly without replacement: every four-ball subset of the twelve individual balls is equally likely.

(a) Find the probability of selecting exactly two red, one blue, and one green ball. (10 points)

Solution. There are $\binom{12}{4}$ equally likely subsets. Choose two of the five red balls, one of the four blue balls, and one of the three green balls:

$$\mathbb{P}(\text{two red, one blue, one green}) = \frac{\binom{5}{2}\binom{4}{1}\binom{3}{1}}{\binom{12}{4}} = \frac{120}{495} = \frac{8}{33}.$$

Credit: Uniform denominator: 3; red choice: 2; blue and green choices: 2; multiplying the counts and forming the ratio: 3. No numerical simplification needed.

(b) Find the probability of selecting at least one green ball. (10 points)

Solution. There are nine non-green balls. The complementary event selects all four balls from these nine, so

$$\mathbb{P}(\text{at least one green}) = 1 - \frac{\binom{9}{4}}{\binom{12}{4}} = 1 - \frac{126}{495} = \frac{41}{55}.$$

Credit: Complement event: 3; count $\binom{9}{4}$: 3; denominator: 2; subtracting the ratio from 1: 2. Accept an equivalent direct count.

5. A density, its mean, and its variance **25 points**

A continuous random variable X has probability density

$$f(x) = \begin{cases} cx^3, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant. Show the integrals you use.

(a) Find c. (5 points)

Solution. A density must integrate to 1:

$$1 = \int_{-\infty}^{\infty} f(x)\,dx = c\int_0^1 x^3\,dx = c\left[\frac{x^4}{4}\right]_0^1 = \frac{c}{4}.$$

Thus $c = 4$, which also makes f nonnegative.

Credit: Normalization integral: 2; integral value: 2; $c = 4$: 1.

(b) Compute $\mathbb{E}[X]$. (8 points)

Solution. Using $f(x) = 4x^3$ on $[0, 1]$,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx = 4\int_0^1 x^4\,dx = 4\left[\frac{x^5}{5}\right]_0^1 = \frac{4}{5}.$$

Credit: Moment integral with the extra factor x: 4; antiderivative and limits: 3; result: 1. Carry forward an earlier value of c.

(c) Compute $\mathbb{E}[X^2]$, and hence $\mathrm{Var}(X)$. (12 points)

Solution.

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx = 4\int_0^1 x^5\,dx = 4\left[\frac{x^6}{6}\right]_0^1 = \frac{2}{3}.$$

Consequently,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{50 - 48}{75} = \frac{2}{75}.$$

Credit: Second-moment integral: 4; evaluating it: 3; variance identity: 3; substitution and arithmetic: 2. Accept a correctly evaluated direct variance integral as an alternative.

6. Collecting four toy types **20 points**

Each cereal box contains one toy. There are four toy types, each equally likely, and different boxes have independent contents. Duplicates are possible. Starting with no toys, let T be the number of boxes opened up to and including the first box that completes the collection of all four types.

(a) After exactly k distinct types have been collected, let p_k be the probability that the next box contains a new type. Find p_0, p_1, p_2, p_3, and explain your reasoning. (6 points)

Solution. Exactly $4 - k$ of the four types are new, so $p_k = (4 - k)/4$. Thus

$$p_0 = 1, \qquad p_1 = \frac{3}{4}, \qquad p_2 = \frac{1}{2}, \qquad p_3 = \frac{1}{4}.$$

Credit: Counting the missing types and dividing by four: 2; four correct probabilities: 1 each.

(b) Let W_k be the number of additional boxes needed to go from exactly k collected types to $k+1$ collected types, including the box containing the new type. Find $\mathbb{E}[W_k]$ for $k = 0, 1, 2, 3$. (8 points)

You may use: A geometric waiting time that counts trials through the first success, with success probability p, has mean $1/p$.

Solution. Until a new type appears, the set of missing types stays the same. Independence of boxes makes this a geometric wait with success probability p_k. For $k = 0$, the first box always supplies a new type. Therefore

$$\mathbb{E}[W_0] = 1, \qquad \mathbb{E}[W_1] = \frac{4}{3}, \qquad \mathbb{E}[W_2] = 2, \qquad \mathbb{E}[W_3] = 4.$$

Credit: 2 points per correctly obtained mean. Carry forward earlier probabilities when the reciprocal rule is applied consistently.

(c) Express T in terms of W_0, W_1, W_2, W_3 and compute $\mathbb{E}[T]$. State the expectation rule you use. (6 points)

Solution. The four successive stages partition all boxes opened, so $T = W_0 + W_1 + W_2 + W_3$. Linearity of expectation gives

$$\mathbb{E}[T] = \sum_{k=0}^{3} \mathbb{E}[W_k] = 1 + \frac{4}{3} + 2 + 4 = \frac{25}{3}.$$

This is an expected number of boxes, not a probability, so it need not be an integer. Independence of the W_k is not needed to apply linearity.

Credit: Stage decomposition: 2; linearity and the sum of stage means: 2; evaluation: 2. Do not require a proof of independence of the waits.

End of paper. Check that you have answered all six problems.